

Department of Computer Science & Engineering
Faculty of Engineering and Information Technology

ASSIGNMENT –I

ODD SEMESTER, 2026-2027

Program Name: B.Tech CSE

Year/Semester:3/5

Course Name: Full Stack Development- Backend

MM:10

Course Code: CS336

- 1.Explain the role and importance of backend development in a web application. Describe how the backend communicates with the client and database. **[BT-2, CO-1, PO-10]**
2. Explain the basic concepts of Node.js and Express.js. Describe the steps required to set up a simple Express.js server. **[BT-2, CO-1, PO-5]**
3. Apply the principles of RESTful API design to develop an API for managing student records. Specify suitable HTTP methods and API endpoints for basic operations. **[BT-3, CO-1, PO-5]**
4. Apply MongoDB CRUD operations to a student collection. Describe the appropriate MongoDB operations for inserting, retrieving, updating, and deleting student records. **[BT-3, CO-2, PO-5]**
5. Analyze the differences between SQL and NoSQL databases with respect to data structure, schema flexibility, scalability, and suitable application scenarios. **[BT-4, CO-2, PO-5s]**